

Frontier AI for Actuaries.

AI helps actuaries perform complex, multi-step actuarial tasks with every step documented and reproducible.

THE WORKBENCH

One collaborative workspace for pricing, reserving, and filing

For all actuaries at admitted and specialty carriers, MGAs, and reinsurers.



Actuarial work is not one workflow. It is a web of intricate, connected tasks.

The more steps an actuary delegates to AI, the less certain they can be of the output. Tesora enables full tracing, documentation, and audit, so the actuary remains in control.



Philo Bishay ✓ • 1st

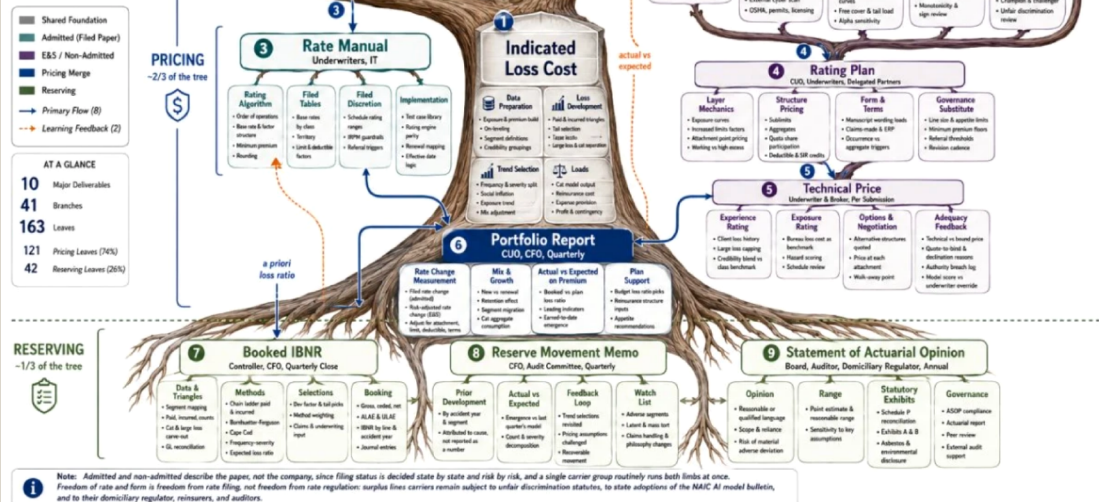
Cofounder & Head of Product @ Tesora | Actuarial | Wharton MBA

2w • 📍

Every actuary dreads one question, "What do actuaries do?" I am happy to say we are no closer to an answer, but if you want to avoid further questions, send this:

What Actuaries Deliver

Two questions. All decisions.
What do we charge?
What do we already owe?
Actuaries own both.



Federico Reyes Gómez and 675 others

23 comments • 28 reposts

2 core domains

Pricing and reserving. Every deliverable rolls up into one of them.

10 major deliverables

The rate filing, the rating plan, the booked IBNR, the statement of actuarial opinion, and six more.

163 individual workflows

Each with its own inputs, its own method, its own order, and its own sign-off.

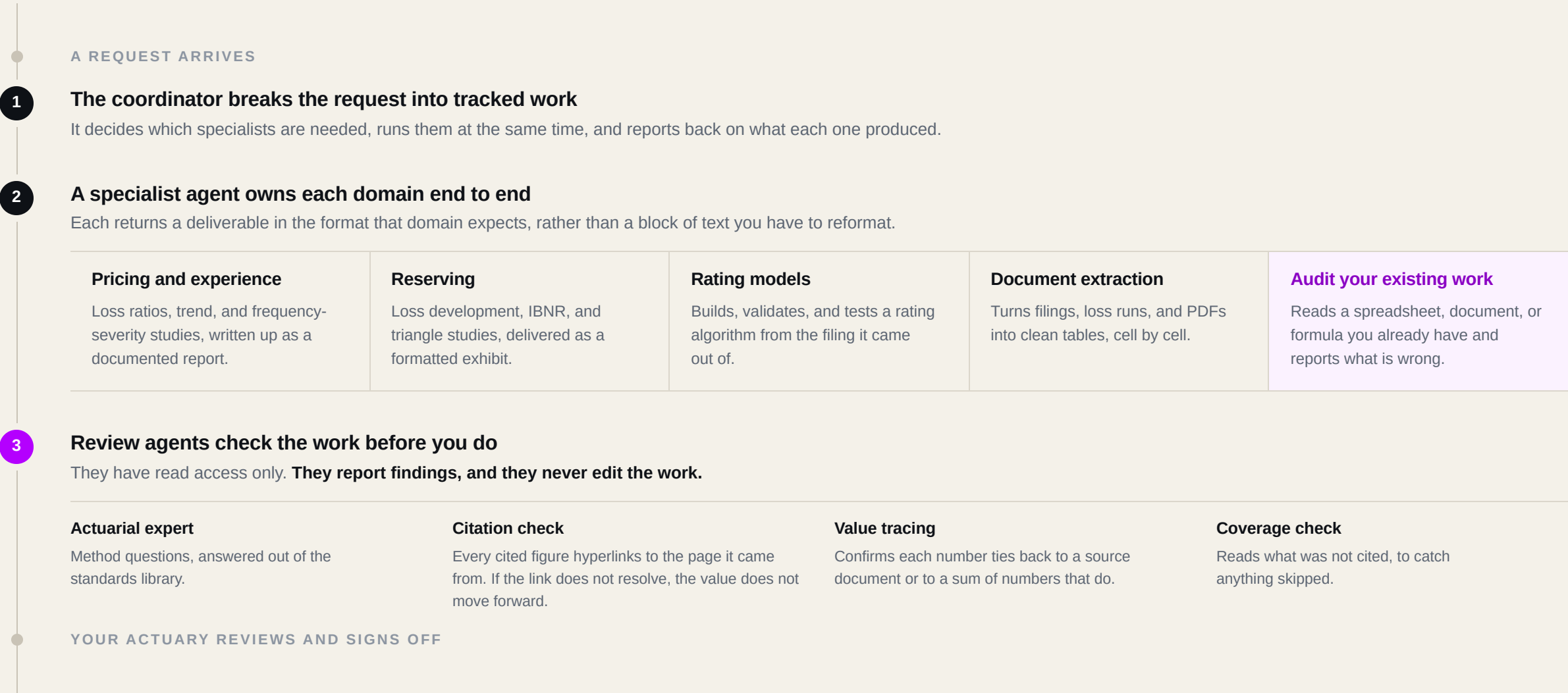
- The judgment lives between these as much as inside any one of them.
- Firm context takes years to build in a person, and a general model arrives with none of it.
- So we start from the methods, the ASOP standards, and the CAS curriculum, then encode an opinionated layer shaped to how your team works.



One agentic harness across all of it

A system of agents keeps every task linked and every value cited, from ingest to filing. That orchestration is the hard part to build, and the harder part to replicate.

No single agent does the whole job, and none of it reaches you unchecked.





Rate changes in **days**, not months.

Source a rater from a SERFF filing, an Excel file, or a script. Tesora rebuilds it, tests it, and deploys it whenever underwriting is ready.

BEFORE: BY HAND

4-6 months

From a requested change to a filed, live rater, and where the months actually go

- | | | |
|----|---|-------|
| 01 | Spec the change
Actuary writes up the revised rates and factors | DAYS |
| 02 | IT ticket, then the queue
The change waits for developer capacity | WEEKS |
| 03 | Re-code the rater
Engineering reimplements the logic by hand | WEEKS |
| 04 | Rebuild exhibits, 50 states
Each state reformatted by hand for SERFF | WEEKS |
| 05 | Manual QA
No test suite, so regressions are caught by eye | WEEKS |
| 06 | Compliance sign-off
Actuarial and legal review the filing | DAYS |
| 07 | File and answer objections
Back and forth with each department of insurance | WEEKS |
| 08 | Release window
Deploy only in the next scheduled window | WEEKS |

AFTER: ON TESORA

1-2 days

Deploy as a web interface or API.

YOU GIVE A plain-language change and the source rater, from SERFF, Excel, or a script.

YOU GET A tested rater with a working GUI, sample ratings, and a versioned API.

← Back to list Version 23 DEVELOPMENT

Inputs Computations Memo Review (23) Deployment

GUI Sample Ratings (15) Governance API

GUI template is valid

Georgia Commercial Auto — Trucks/Tractors/Trailers

Liability + Physical Damage rating with zone-rated Long-Distance segment

Vehicle & Classification

Territory *
101

Size Class *
Heavy Trucks

Business Use *
Commercial

Radius *
Local

Secondary Class *
Not Otherwise Specified — Other

Fleet *
No (Non-Fleet)

Final Premium

\$3,610.00

The triangle is squared before you sit down.

A quarterly close moves between actuarial, IT, and finance, and most of the calendar goes to assembling the data rather than judging it.

BEFORE: BY HAND

~2 weeks

From close to a booked IBNR, and who is waiting on whom

01 Request the data

Actuarial files a ticket for the quarter's extract

IT

02 Wait on the extract

The pull sits behind other reporting work

DAYS

03 Reconcile to the ledger

Paid and incurred tied back to finance, by hand

FINANCE

04 Rebuild the triangles

Segment mapping and large-loss carve-outs redone each quarter

DAYS

05 Select factors and methods

Chain ladder, Bornhuetter-Ferguson, Cape Cod, tail

ACTUARIAL

06 Rebuild the exhibits

Roll-forward and movement memo reformatted by hand

DAYS

07 Peer review

A second actuary retraces the work from scratch

DAYS

08 Book and explain

Answer the CFO, the auditor, and the board

DAYS

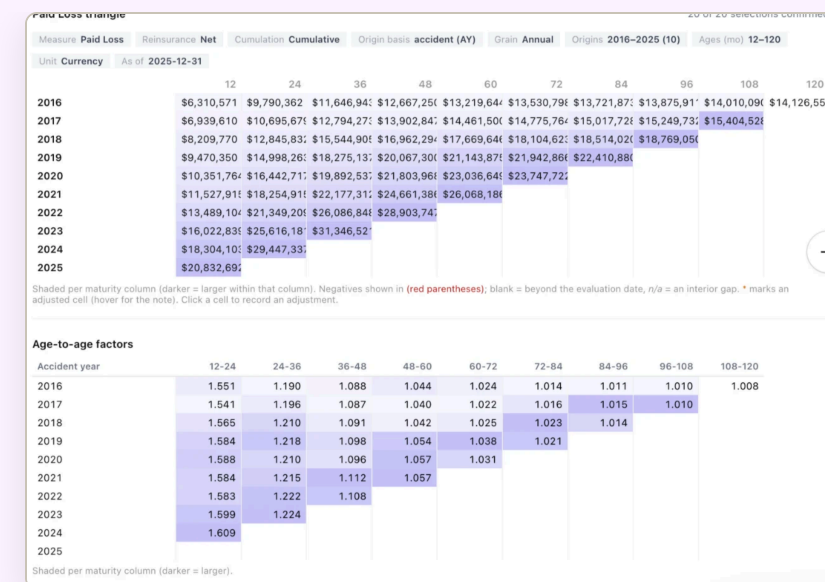
AFTER: ON TESORA

3-4 hours

Triangles squared on ingest, so the quarter starts at the selection

YOU GIVE The loss run, as it comes out of the system.

YOU GET Reconciled triangles, method comparisons, and an IBNR with every value traced to source.



Rebuild what a competitor charges from their filings, then **price your book through it.**

A filing gives you tables, rules, and inputs rather than a working model, and the order of operations is rarely spelled out. Turning that into something you can run your own risks through is the work, and by hand it takes long enough that the competitor has often filed again before you finish.

BEFORE: BY HAND

2-3 weeks

From a competitor filing to a usable read, and where it goes wrong

- | | | |
|----|--|------|
| 01 | Pull the filings
An analyst searches SERFF, state by state | DAYS |
| 02 | Read the rate manual
Rules, factors, and order of operations, by eye | DAYS |
| 03 | Rekey the rate tables
Hundreds of tables retyped into a spreadsheet | DAYS |
| 04 | Rebuild the algorithm
The logic is inferred, then argued over | DAYS |
| 05 | Price a sample
A dozen test risks, never the real book | DAYS |
| 06 | Compare by hand
Deltas built in a one-off workbook nobody owns | DAYS |
| 07 | Explain the gaps
Each difference traced back to a single factor | DAYS |

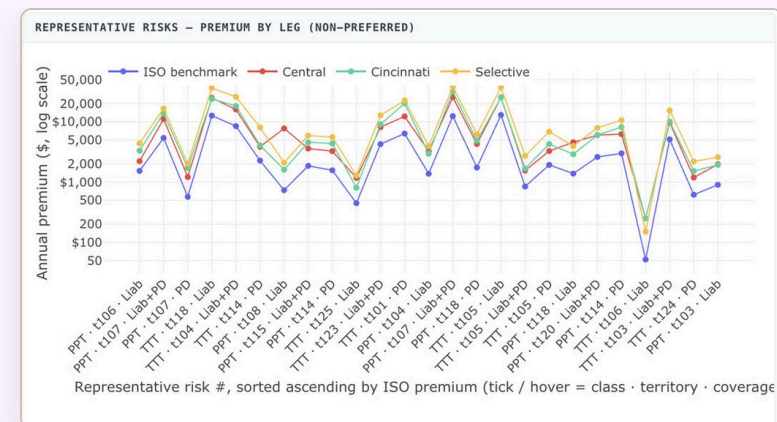
AFTER: ON TESORA

10-15 minutes

An executable competitor rater, and your whole book priced through it

YOU GIVE A competitor, a state, and a line. Or the SERFF filing itself.

YOU GET Their rating logic running against your risks, every factor cited to the page it came from.





The full context is baked into the work.

Every session holds the data it used, the assumptions behind it, and the reasoning that got there. Reopen it years later and nothing has to be reconstructed, because the context was never separate from the answer.

Rate Rule Form

Profit & Contingencies	7.0%
Total expense & profit load	47.0%
Expected Loss & LAE Ratio (ELR)	53.0%
Projected Ultimate Loss Ratio	54.7%
Adequacy margin (ULR - ELR)	1.7% short

Read: the thin spot is the **47% load** (30 pts of it commission), and the book runs ~1.7 pts past its permissible loss ratio — a modest inadequacy with no cushion.

Caveat (carried from the exhibit): ILFs are relativities, not dollar prices; no competitor base rates or retention/deductible tables exist in the folder. Preliminary analyst work product, subject to credentialed-actuary review.

The full CSV is downloaded at [CIC_vs_Peers_ILF_Competitive_Benchmark.csv](#), and the narrative report is [CIC vs Cyber Peers — ILF & Competitive Benchmark ...](#). Want me to render this as a styled Excel exhibit, or open the report next?

bernegger-2024-integrated-experience-exposure-rating.pdf

Download

Origin: User upload
Type: application/pdf
Size: 1.5 MB
Status: pending
Uploaded: Aug 7, 2026, 06:57 PM
Document ID: DOC-5c0c8dfe-1a3a-4eeb-8c76-5ab1947ac4dd
File ID: 54238c79-b072-45c7-92cb-096c7b81f242

SUMMARY

No summary available. The ingestion workflow either hasn't finished or didn't produce one for this document.

Generic framework for a coherent integration of experience and exposure rating in reinsurance

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Received: 7 October 2023; Revised: 11 April 2024; Accepted: 12 April 2024; First published online: 27 August 2024

Keywords: Experience rating; exposure rating; short tail; long tail; generative model; Bayesian inference; reduced statistics; model calibration; Monte Carlo simulation

Abstract

This article introduces a comprehensive framework that effectively combines experience rating and exposure rating approaches in reinsurance for both short-tail and long-tail businesses. The generic framework applies to all non-life lines of business and produces comprehensive nonoperational treaty business. The approach is based on three pillars that enable a coherent usage of all available information. The first pillar comprises an exposure-based generative model that emulates the generative process leading to the observed claims experience. The second pillar encompasses a standardized reduction procedure that maps each high-dimensional claim object to a low-dimensionality coupled reduced random variable. The third pillar comprises calibrating the generative model with retrospective Bayesian inference. The derived calibration parameters are fed back into the generative model, and the reinsurance contracts covering future cover periods are rated by projecting the calibrated generative model to the cover period and applying the future contract terms.

1. Introduction

There are two approaches to probability theory: the first is the frequentist interpretation, also known as standard, orthodox, or classical statistics, and the second is the Bayesian interpretation (DeGroot et al., 1994; Lee, 2012; Gelman, 2013; Klugman et al., 2019; Murphy, 2022, 2023). This distinction also applies to the various statistical and modeling approaches.

The Bayesian interpretation has multiple advantages, but frequentist approaches are still popular and widely used for two main reasons. First, they are computationally 'cheap' compared to Bayesian methods, and second, they do not presuppose knowledge regarding priors. Bayesian inference is, nevertheless, widely applied in various domains of science (Parr et al., 2022; Palmer, 2022; Harrison et al., 2023), and it has become the foundation of machine learning (Murphy, 2022, 2023). Bayesian approaches are also gaining popularity in disciplines like insurance and reinsurance, which have traditionally been dominated by frequentist approaches (Bühlmann and Straub, 1970; Bühlmann and Gisler, 2005; Zhang et al., 2012; Klugman et al., 2019; Goffard et al., 2023).

Reinsurance companies face multiple challenges when using models for assessing the assumed risks and for taking business decisions like risk selection, transactional pricing, reserving, investing, and corporate risk management, and they rely on a variety of methods and tools for managing their risks (Antal, 2009; Albrecher, 2017; Müllenshahn, 2022). The actuarial models implemented by a reinsurer are based on simplified assumptions, and reinsurers know that the underlying assumptions are rarely fulfilled in the real world. One way to deal with the issue is to complement experience-based rating models with exposure-based models. The latter can incorporate all relevant features of the generative process that affect the rating. All such features must be calibrated, which is accomplished with market data. However, cedent-specific features (hidden variables) must also be reflected in the rating of reinsurance contracts.

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Implementation

The models are wired into your loss runs, policy systems, and ledger, running inside your own environment.

Governance

Who ran what, on which data, and who signed it. Finance and IT find what they need without waiting on an actuary to explain it.

Compliance

Built to ASOP from the start, so a filing survives the objection, the examination, and the auditor years later.

Continuity

When an actuary leaves, their assumptions, sources, and reasoning stay in the session. The next person picks up the thread.



What's the best way to **implement AI**?

Your implementation of AI today will dictate its value, translatability, and reproducibility going forward.

	Actuarial consultants Milliman, WTW, Oliver Wyman	Your team with AI In-house, on ChatGPT or Claude	Actuarial software hyperexponential, Earnix, Akur8	T Tesora Frontier models, in your format
Turnaround	✗ Weeks to months, in their queue	≈ Quick drafts, still checked by hand	✓ Fast and repeatable once it is live	✓ Minutes to days, in your workflow
Setup & fit	≈ A fresh engagement scoped each time	✓ No one knows your book better	✗ Heavy implementation on their platform	✓ 60-day build in your VPC, on your own data
Your raters & filings	✗ Generic deliverables in their template	≈ Your format, rebuilt by hand each time	≈ Rebuilt inside their system, their way	✓ Your raters, versioned and ready to deploy
Source & audit	✓ Independent, credible workpapers	✗ No source to trace, and the answer drifts	≈ Logged inside the platform	✓ Cell-level citations, append-only log
Scope	≈ Whatever you scope and pay for	≈ One step at a time, gaps in between	≈ Pricing or reserving, rarely filing too	✓ Pricing, reserving, filing, and close in one place



Strength



Partial or conditional



Gap

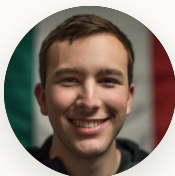


A team built at the intersection of frontier AI research and decades of actuarial experience, **the two disciplines this problem demands.**

**Vivek Rao**

CEO

Former private-equity investor and McKinsey consultant focused on P&C insurance.

**Federico Reyes**

CTO

Stanford AI researcher. Graph neural nets and relational data at Kumo AI, acquired by Nvidia.

**Philo Bishay**

CPO & CHIEF ACTUARY

Former chief actuary at W. R. Berkley E&S. Capital modeler at Gen Re; built and sold a brokerage.

**Michael Palmich**

HEAD OF SALES

25+ years in the actuarial software space, most recently at Akur8 and Milliman.

**Eric Lam**

FOUNDING ENGINEER

MIT for CS, physics, and math, with **three actuarial exams** done.

ADVISORY BOARD

**Ethan Genteman**

FCAS, MAAA

Chief Actuary, Palomar (NYSE: PLMR)

**Mario DiCaro**

FCAS, CERA

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**Joanne Chen**

General Partner, Foundation Capital

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Leading AI investors, backing enterprise founders since 1995

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The accelerator behind Stripe and Airbnb

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Above all, we want to help you **win.**

We are a thought partner first. This technology is powerful, and it only helps once it is opinionated: built with a real point of view on how the work gets done and defended.

We are here to push the frontier forward with the right partners, not to sell you a license. A law firm does the legal work rather than selling software to lawyers, and we hold ourselves to that standard: **we take ownership of the outcome, not just the tooling.** Every deployment starts with a proof of concept anchored on your own success metrics, and we work with your team to find where the workbench sets the shared standard first.

Let's set up a call

sales@tesora.ai

DE-RISKED BY DESIGN

- ✓ It runs in your own environment. We never hold your data.
- ✓ SOC 2 Type II, with the highest data-security controls.
- ✓ We clear your security review before any work touches a filing.

Appendix



Per actuary, the recurring month goes from 150 hours to about 60.

A full-time actuary has roughly 160 hours in a month, and most of them go to recurring work. Tesora compresses the mechanical grind, the data pulls, the reconciliations, the model updates, and hands back **upwards of 90 hours to every actuary**, to spend on judgment instead of assembly.

Where one actuary's recurring month goes		reclaimed	still hands-on	ILLUSTRATIVE · MODELED
Data extraction & reconciliation		39	→ 4	10×
Pricing model updates		28	→ 7	4×
Loss trend analysis		23	→ 12	2×
Reserve & loss development		21	→ 12	1.7×
Management & board reporting		17	→ 10	1.7×
Reinsurance program analysis		12	→ 4	3×
Peer review & documentation		10	→ 8	1.3×
Monthly total, per actuary		150	→ ~60	2.5×

~90 hours back per actuary, per month

A 2.5× blended throughput gain across the recurring month, with the same team and the same sign-off.

Smarter, not harder

Time freed for strategy and selections rather than assembly.

Unlock your actuaries

More products, more territories, and more actuarial capability from the same team.

Hours are per actuary and reflect a typical ~150 hour recurring workload within a roughly 160 hour month, with per-workflow gains from Tesora's engagement framework. Figures are modeled and confirmed against measured results during a build.



We run on top of Claude, and we are model agnostic.

The model is the engine, not the product. What an actuary needs around it is infrastructure: the data wired in, the workflow decided, the cost controlled, and the record kept. That is the part we build, and we build it for you.

We guide the model

A raw model will answer anything, confidently. Ours is held to an actuarial method: which factors to consider, which checks to run, when to stop and ask. On a loss trend that means frequency and severity are fit separately and tested against your own history before anything is selected.

We wire the infrastructure

Your loss runs, policy systems, filings, and ledger are connected once and stay connected. A chat window starts empty every time. Ingest once here, and every study after it reads the same reconciled book.

The workflow is opinionated

Reserving and pricing have a right order of operations, and we encode it. Chain ladder, Bornhuetter-Ferguson, and Cape Cod run side by side, so the selection stays a choice your actuary makes rather than whatever the prompt happened to produce.

We manage the cost

Running a real book through a frontier model naively is slow and expensive. Extraction runs on a small model and the judgment calls go to the strongest one, so the economics work at production scale.



We exist to build **your actuarial infrastructure**, not to serve the models or ourselves. When a better model ships, you get it. What stays is the system we built around your book.



You buy the outcome. We handle the models.

We make sure the work runs on the best model for it and lands compliant, so model efficiency is our problem rather than a line on your budget. That is a job the model vendors cannot do for you, because every one of them grows revenue by selling you more of what they sell.



The most efficient model per outcome, at the right cost

Each step is matched to the model that delivers it correctly for the least spend. Extraction runs small, judgment runs strong, and you are billed for the result rather than the tokens behind it.



Our incentives point the other way

A model vendor is never going to tell you to buy less of what it sells. We are not paid by the token, so **reducing your spend is work we are free to do.**



Your infrastructure moves to whichever model is best

Your workflows, your data wiring, and your audit trail live with us, so a stronger model arrives without a migration, a rebuild, or a new vendor.



We keep learning your systems

The longer we work with your team, the more of your data, formats, and quirks the workbench encodes, and that knowledge carries across every upgrade.



Audits matter more as the models do more

Better models take on more complex work, which raises rather than lowers the bar on proving it. Every value stays cited and reproducible, and **the advantage compounds for whoever moves first.**